



Advanced Digital Image Processing Lecture #3

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Summary of Previous Lecture

- ❖ Definition of Digital Image
- ❖ Block Diagram of Image Processing Sequences
- ❖ Digitization and Quantization
- ❖ Common Values of Digital Image Parameters
- ❖ Characteristics of Image Operations
- ❖ Applications of Digital Image Processing

Image Representation And Modeling (1)

- ❖ Image could be represented by luminances of an object in a scene (Picture taken by ordinary camera)
- ❖ The absorption capabilities of the body tissue (X-ray imaging)
- ❖ The Temperature Profile of a region (Infrared imaging)
- ❖ The radar cross section of a target (Radar imaging)
- ❖ The Gravitational field in an area (Geophysical imaging)

Image Representation and Modeling (2)

- ❖ An Image is represented by 2-D function

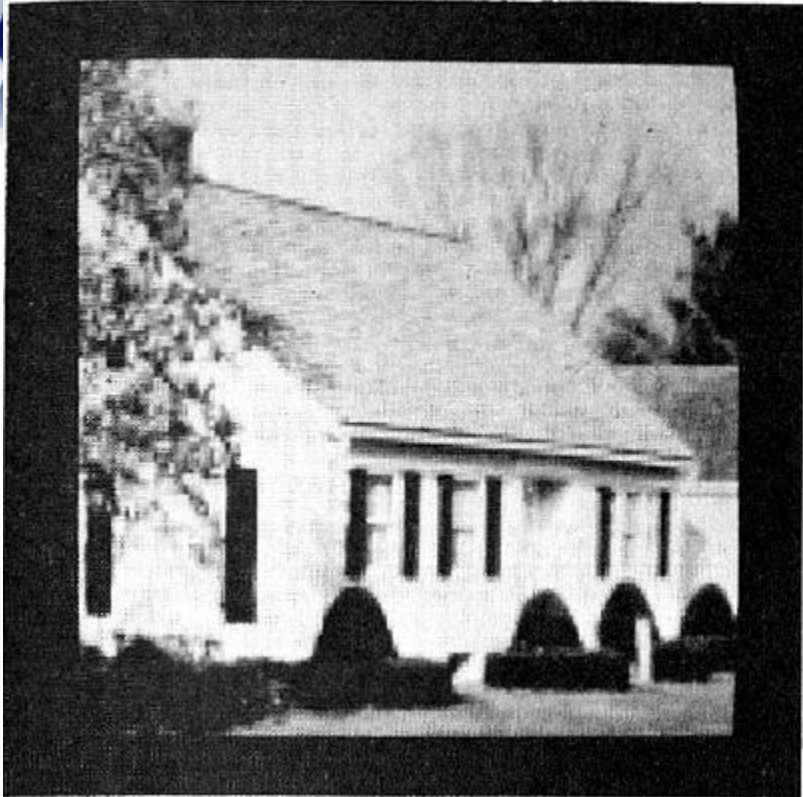
$$f(\mathbf{x}) = f(x, y)$$

Where, $f(x,y)$ is the brightness of the gray level of the image at spatial coordinate (x,y)

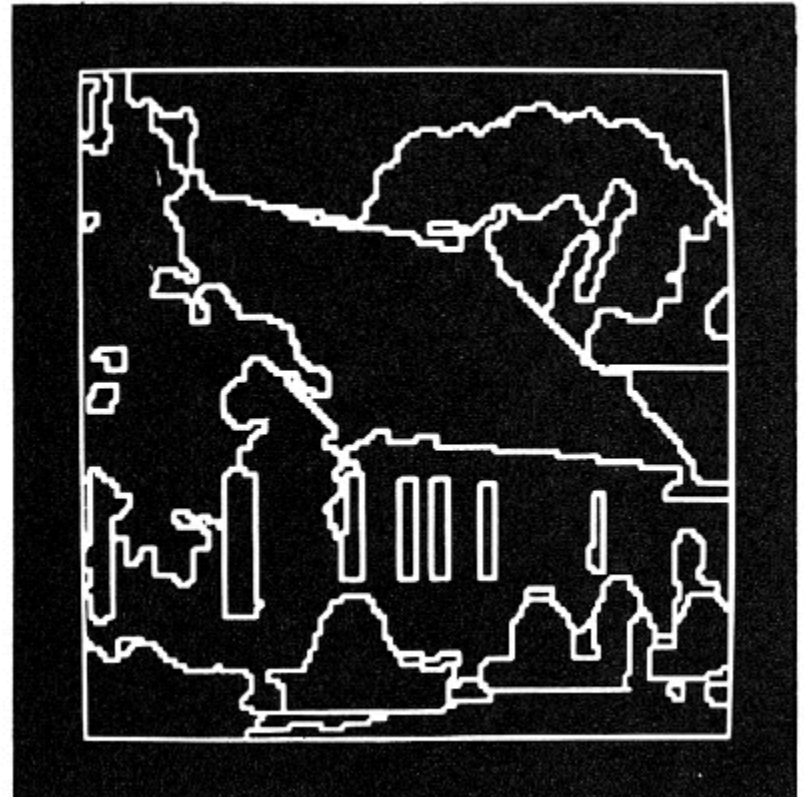
- ❖ Image models gives a logical or quantitative description of the properties of this function,

$$f(\mathbf{x}) = \left\{ f_{\text{red}}(\mathbf{x}), f_{\text{blue}}(\mathbf{x}), f_{\text{green}}(\mathbf{x}) \right\}$$

Example: Image Representation (1)

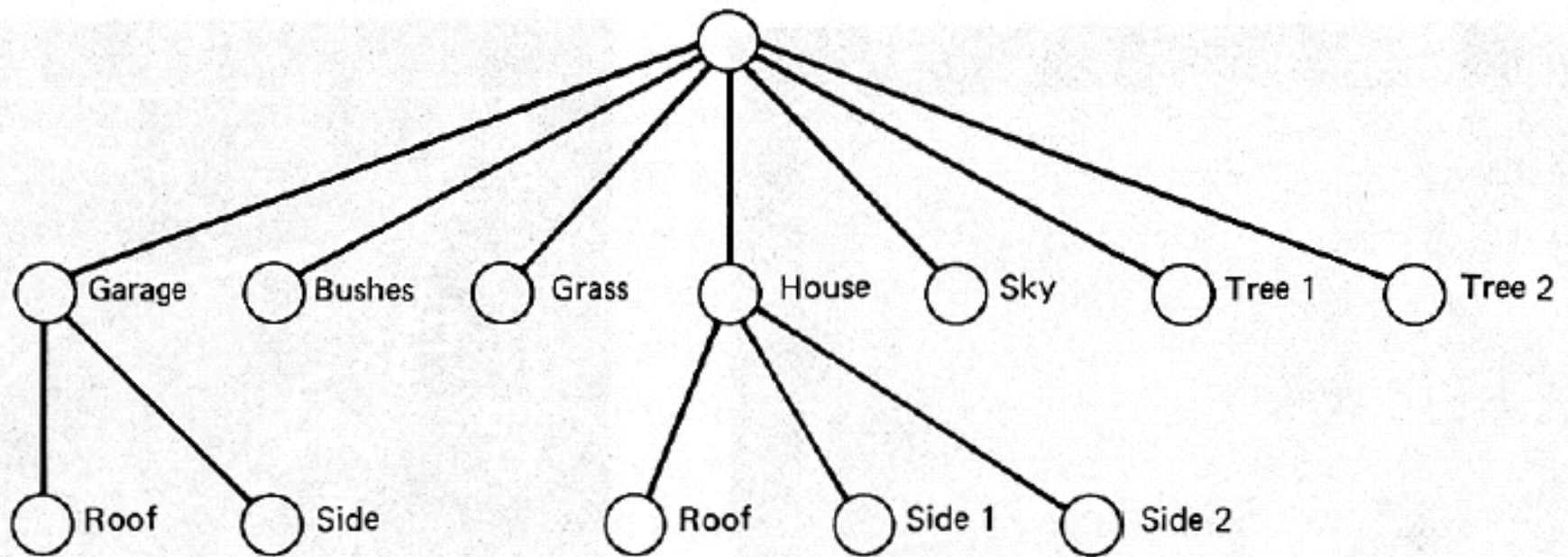


Iconic Representation



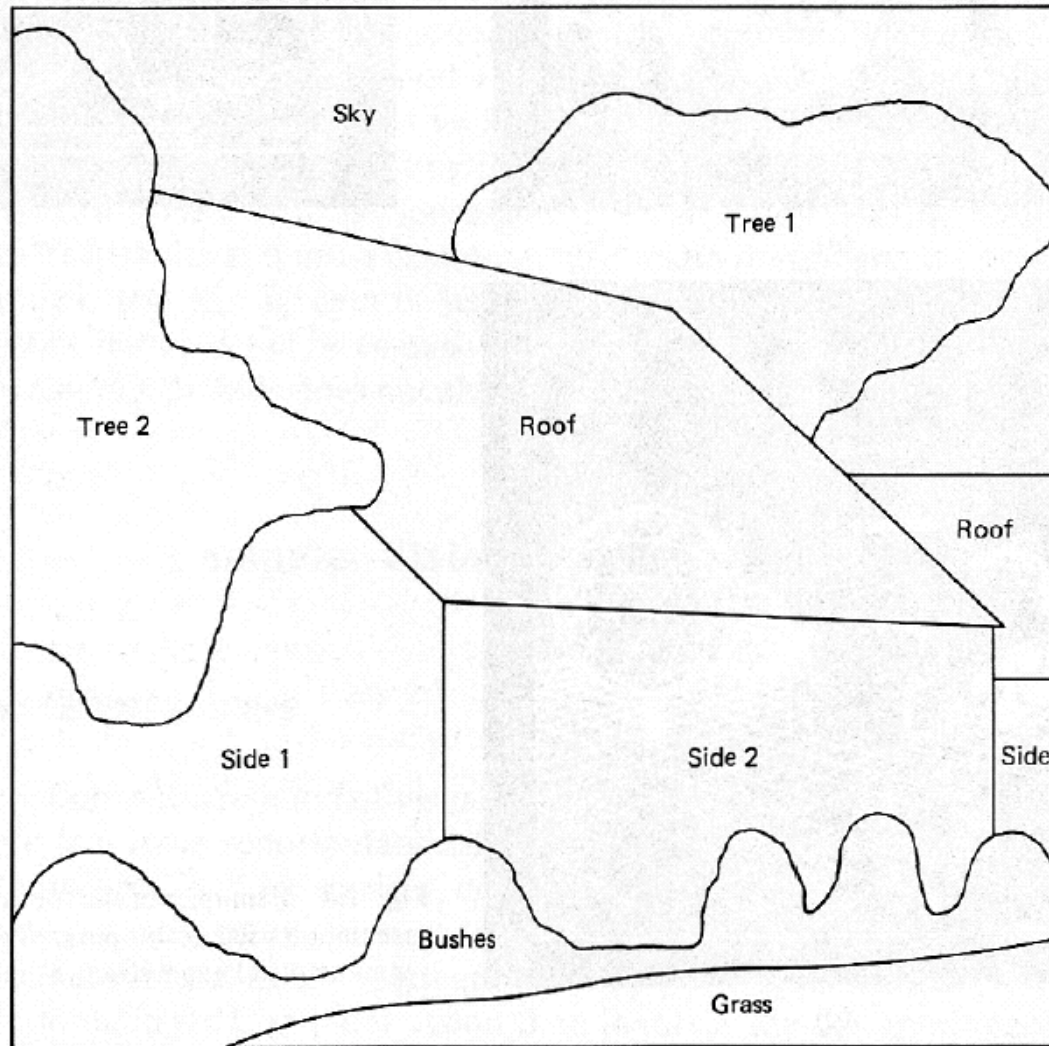
Segmented

Example: Image Representation (2)



Relational (1)

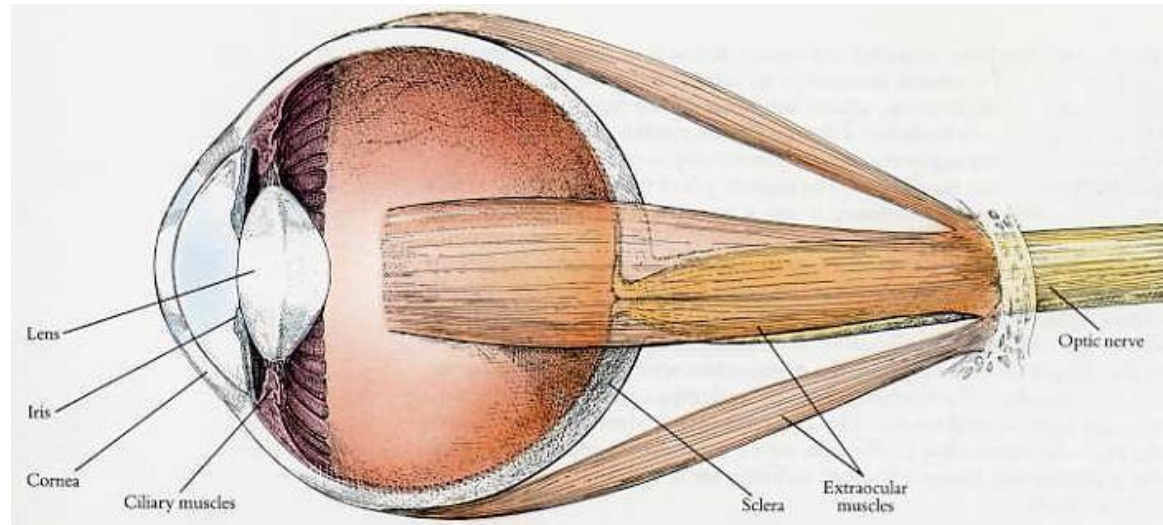
Relational Representation(2)



Mathematical Model

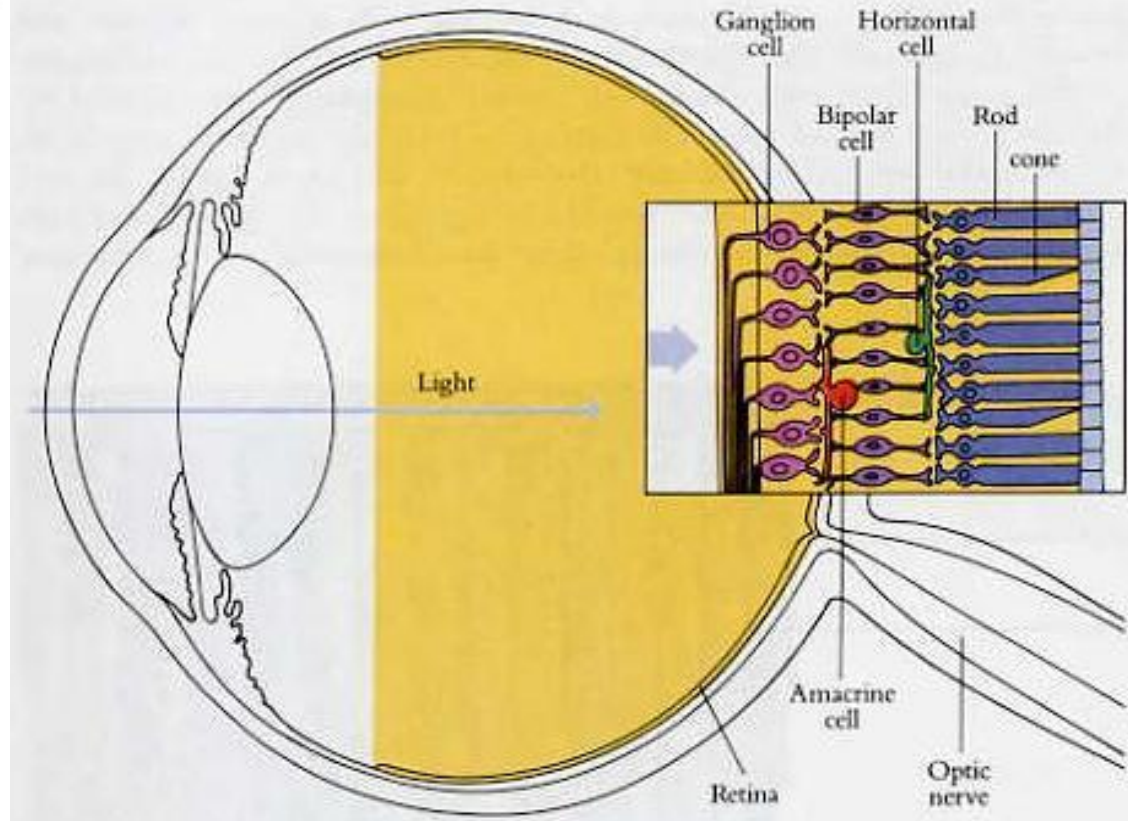
- ❖ An image function is the fundamental abstract of an image
- ❖ A geometrical model describes how three dimensions are projected into two dimensions
- ❖ A radiometrical model shows how the imaging geometry, light sources and reflectance properties of objects affect the light measurement at the sensor
- ❖ A spatial frequency model describes how spatial variation of the image may be characterized in a transfer domain
- ❖ A color model describes how different spectral measurements are related to image color
- ❖ A digitizing model describes the process of obtaining the discrete sample

Anatomy of Human Eye (1)



➤ **The cornea and the lens focus the light rays onto the back of the eye. The lens regulates the focusing for near and far objects by becoming more or less globular.**

Anatomy of Human Eye (2)



➤ the light has to pass through the ganglion-cell and bipolar-cell layers before it gets to the rods and cones.

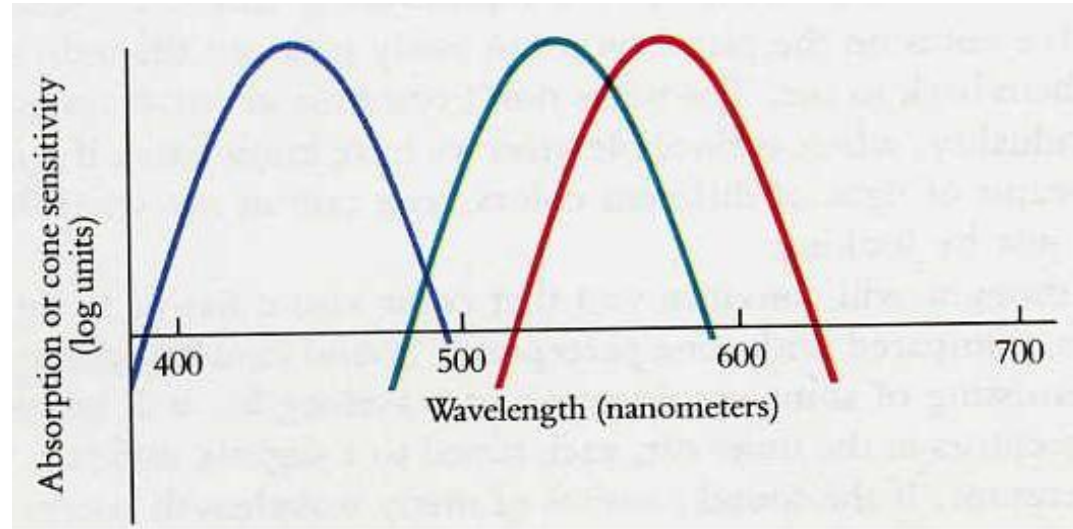
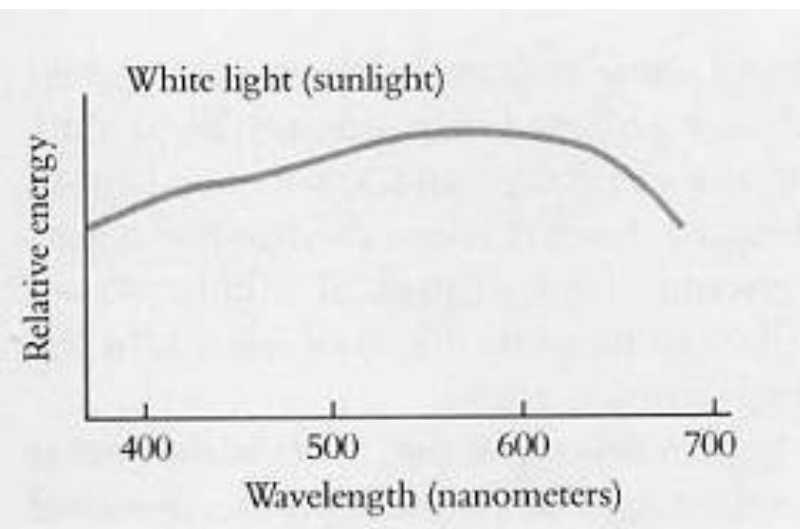
How Human Visual System Works?

❖ Retina of human eye has two types of photoreceptors:

- ❖ Rods: 100 millions, long and thin, stimulate in lower magnitude of illumination (scotopic vision)
- ❖ Cones: 6.5 millions, Short, thick, 5-6 order of illumination (well light, photopic vision)
- ❖ For midlevel vision (mesopic vision) both are activated
- ❖ Fovea: Center of retina densely packed by cones
- ❖ Pupil: For bright light act as a low pass filter (2 mm dia)
- ❖ Cones are laterally connected by horizontal cells and have a forward connection with bipolar cell and ganglion cell
- ❖ Ganglion cell connect with optical nerve

Definition of Light

- ❖ Light consists of particles called photons, each one of which can be regarded as a packet of electromagnetic waves.
- ❖ Our eyes can detect electromagnetic energy at wavelengths between 400 and 700 nanometers.

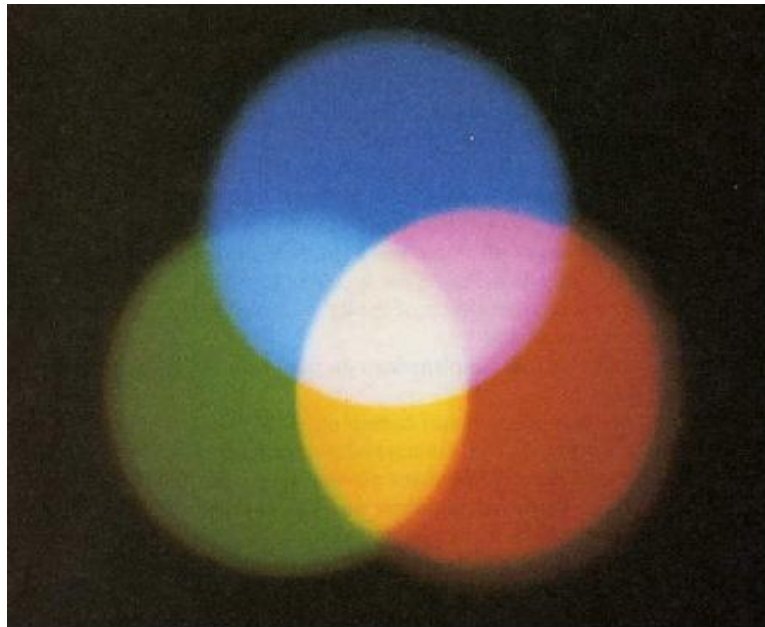




Properties of Light

- ❖ When light hits an object, one of three things can happen:
 - ❖ the light can be absorbed and the energy converted to heat
 - ❖ it can pass through the object
 - ❖ or it can be reflected

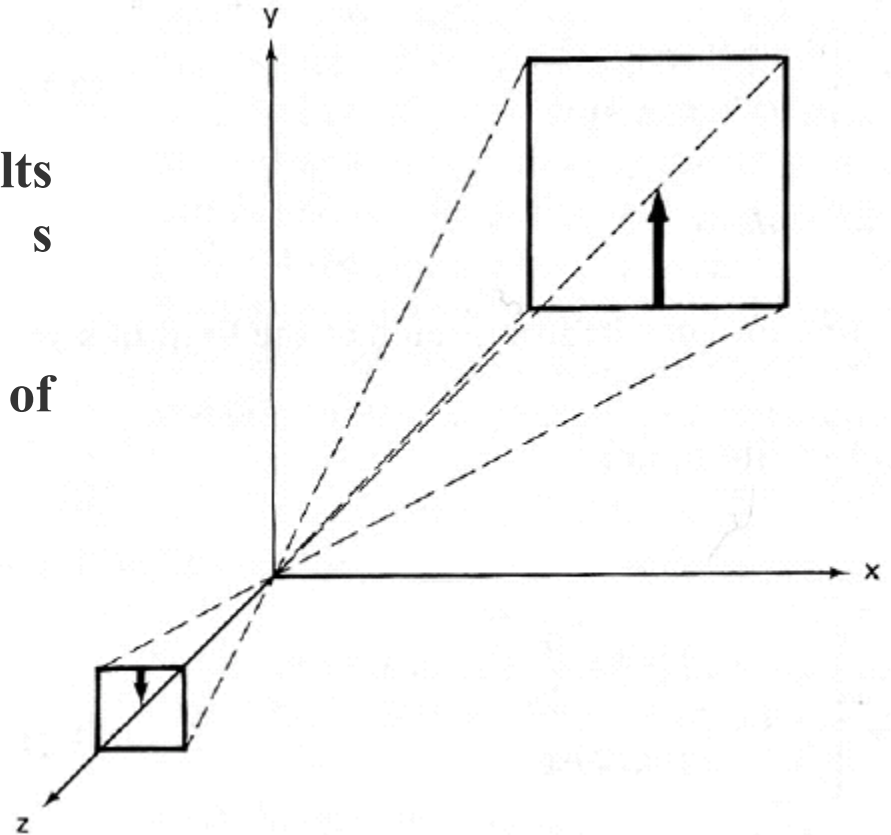
Overlapping of three Basic colors



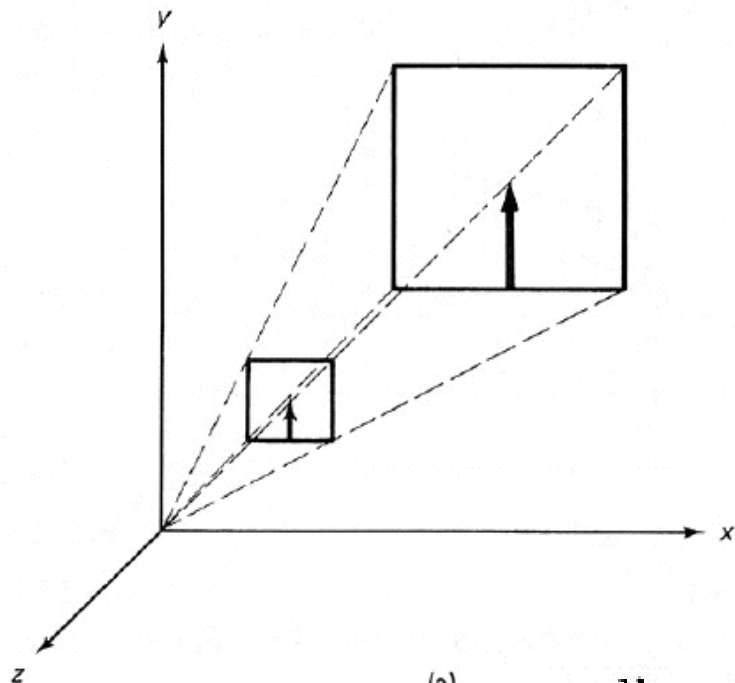
Three overlapping spots (red, green, and blue) are projected onto a screen so that they overlap. Red and green give yellow, blue and green give turquoise, red and blue give purple, and all three—red, blue, and green give white.

A Geometric Camera Model (Monocular Vision)

- ❑ For a pinhole camera the image results from projecting scene points through a single point onto an image plane
- ❑ The image plane is behind the point of projection and the image is reversed

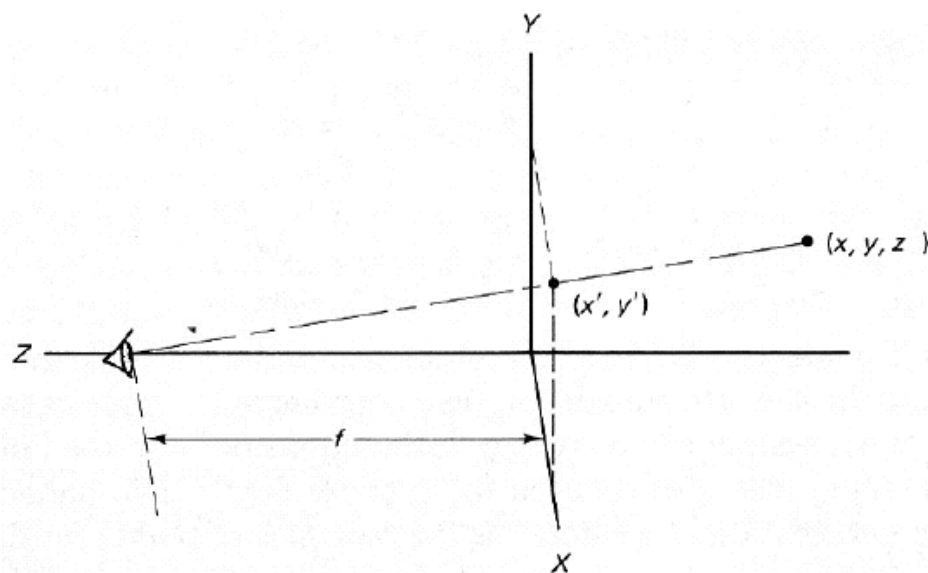


Camera Model (2)



(a)

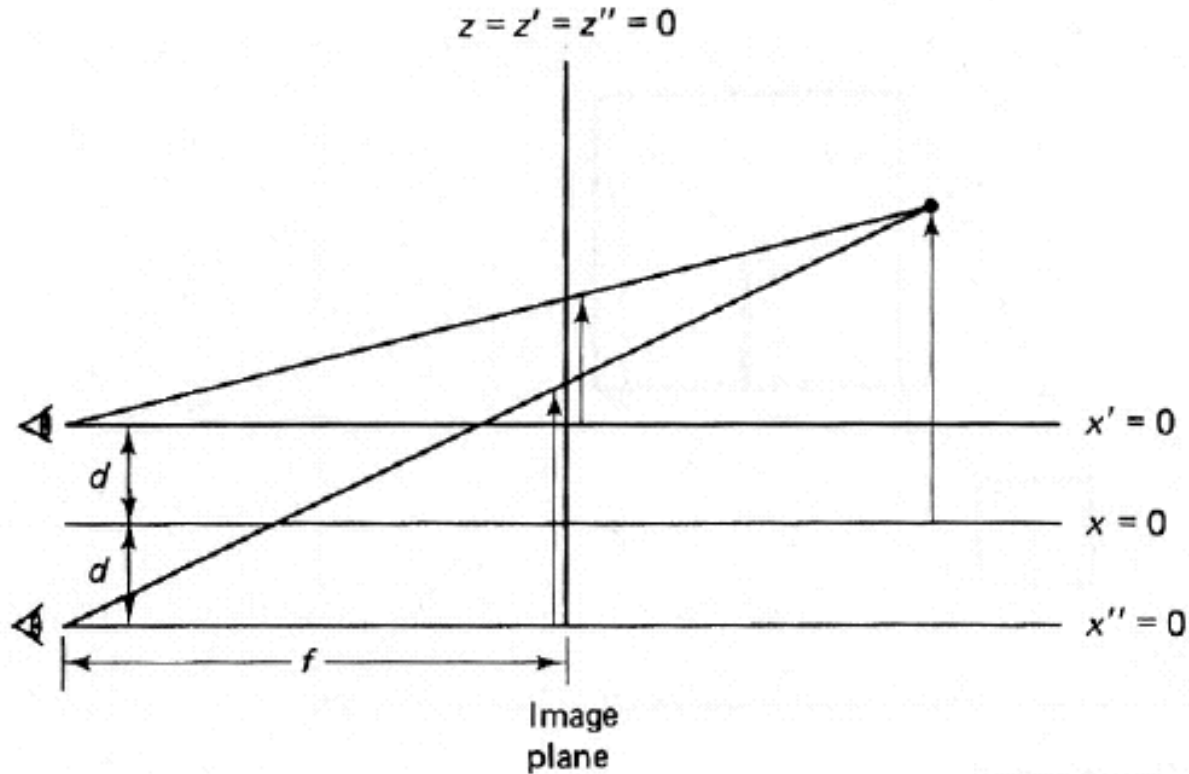
$$\frac{y}{f - z} = \frac{y'}{f}$$



(b)

$$\frac{x}{f - z} = \frac{x'}{f}$$

Binocular Vision



$$x' = \frac{(x - d)f}{f - z}, \quad x'' = \frac{(x + d)f}{f - z}, \quad z = f - \frac{2df}{x'' - x'}$$

Garibaldi Fish



The blue spots on this fish fade as it grows older, disappearing when it matures.



Thanks to all !!!